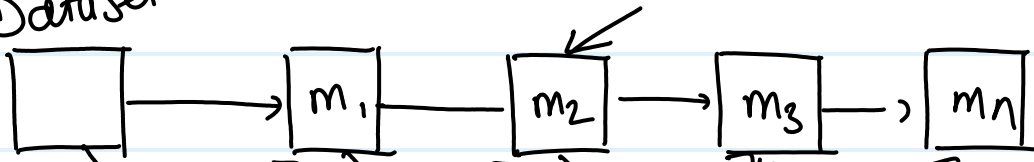


Gradient Boosting.

predicted data (Error)

Dataset

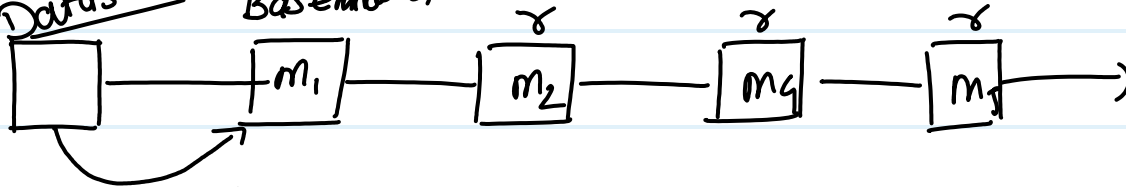


	1	0		
✓	1	1	uniform	1
	0	1		1.2
✓	0	0		0.7
				1
				1.2
				0.7

$$\alpha m_1 + \alpha m_2 + \alpha m_3 =$$

Dataset

Basemodel



Average

y	$\hat{y}$	$(y - \hat{y})$
10	13	-3
✓ 20	18	2
15	13	2
16	13	3

Archer $\rightarrow$

Target = 100 ✓

$\checkmark m_1$	shot - 1 = 70 ✓	$100 - 70$	$= 30$	r
$\checkmark m_2$	shot - 2 = 20	$30 - 20$	$= 10$	
$\checkmark m_3$	shot 3 = 7	$10 - 7$	$= 3$	
$\vdots$				
$\checkmark m_n$	shot n = 2	$3 - 2$	$= 1$	

Final $\Rightarrow 70 + 20 + 7 + 2 = \underline{\underline{99}}$

$$F_1(x)$$

$$F_2(x)$$

$$F_3(x)$$

$$f_+(x) = f_{(t-1)}(x) + \text{learning rate} \times h_t(x)$$

$$\text{Residual} = \text{Actual} - \text{Predicted}$$

$$y - \hat{y}$$

price prediction

Size sqft ^x	price ^y	f_0 $\hat{y}$	residual r_i $(y - \hat{y})$
600	40	79.2	-39.2
800	55	79.2	-24.2
1000	65	79.2	-14.2
<u>1200</u>	72	79.2	-7.2
1400	80	79.2	0.8
1600	85	79.2	5.8
1800	90	79.2	10.8
2000	100	79.2	20.8
2200	110	79.2	30.8
2400	95	79.2	15.8

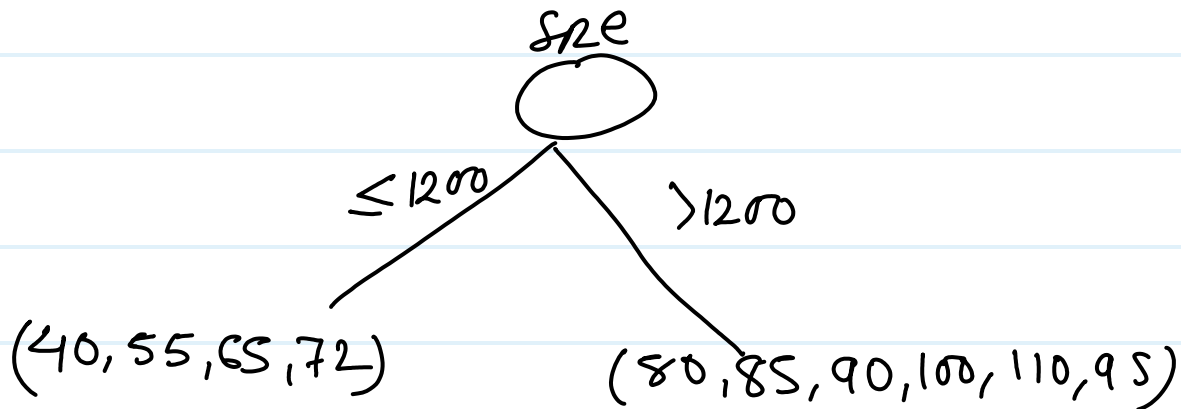
Base model $f_0 \rightarrow$

$$\Rightarrow \frac{(40 + 55 + 65 + 72 + 80 + 85 + 90 + 100 + 110 + 95)}{10}$$

$$\Rightarrow 79.2$$

model F_1 Tree

Tree on residual (split at $sre \leq 1200$)



$$\leq 1200 = \frac{-392 + (-24.2) + (-14.2) + (-7.2)}{4}$$

$$= -21.2$$

$$> 1200 = \frac{0.8 + 5.8 + 10.8 + 20.8 + 30.8 + 15.8}{6}$$

$$= 14.13$$

learning rate $\eta = 0.5$

$$F_1(x) = F_0 + \eta \times h_1(x)$$

f_0	$\eta \times h, x$	f_1	y	σ_2
79.2	$0.5 \times -21.2 = -10.6$	68.6	40	$40 - 68.6 = -28.6$
79.2	$0.5 \times -21.2 = -10.6$	68.6	55	-13.6
79.2	$0.5 \times -21.2 = -10.6$	68.6	65	-3.6
<u>79.2</u>	$0.5 \times -21.2 = -10.6$	68.6	72	3.4
79.2	$0.5 \times 14.13 = 7.07$	86.27	80	-6.27
79.2	$= 7.07$	86.27	85	-1.27
79.2	$= 7.07$	86.27	90	3.73
79.2	$= 7.07$	86.27	100	13.73
79.2	$= 7.07$	86.27	110	23.73
79.2	$= 7.07$	86.27	95	8.73

next free (m_2)

$$\text{size } (\leq 800) = \frac{-28.6 + (-13.6)}{2}$$

$$\boxed{= -21.1}$$

$$\text{size } (> 800) = -3.6 + 3.4 + (-6.27) + (-1.27) + 3.73$$

$$+ 13.73 + 23.73 + 8.73$$

$$\boxed{= 5.28}$$

F_1	$\eta \times h_2(x)$	F_2	y	r_3
68.6	$0.5 \times -21.1 = -10.55$	58.05	40	$40 - 58.05 = -18.05$
68.6	$0.5 \times -21.1 = -10.55$	58.05	55	$= -3.05$
68.6	$0.5 \times 5.28 = 2.64$	71.24	65	$= -6.24$
68.6	$= 2.64$	71.24	72	$= 0.76$
86.27	$= 2.64$	88.91	80	$= -8.91$
86.27	$= 2.64$	88.91	85	$= -3.91$
86.27	$= 2.64$	88.91	90	$= 1.09$
86.27	$= 2.64$	88.91	100	$= 11.09$
86.27	$= 2.64$	88.91	110	$= 21.09$
86.27	$= 2.64$	88.91	95	$= 6.09$

$$\begin{array}{ccc}
 r_1 & r_2 & r_3 \\
 -39. & -28 & -18
 \end{array}$$

$$F_{\text{final}}(x) = F_0 + \eta \times h_1(x) + \eta h_2(x) + \eta h_3(x) + \dots + \eta h_n(x)$$

$$= 79.2$$

$$= 0.5 \times \text{Tree 1 prediction}$$

$$= 0.5 \times \text{Tree 2 prediction}$$

$$\begin{array}{rcl}
 \checkmark & 100 & \\
 m_1 - 70 & = & 30 \\
 m_2 - 20 & = & 10 \\
 m_3 - 7 & = & 3 \\
 m_4 - 2 & = & 1 \\
 m_5 - 0.5 & = & \boxed{0.5}
 \end{array}$$

$$\begin{array}{rcl}
 600 & \left. \begin{array}{c} \cdot 1 \\ \left[\begin{array}{c} 1 \\ 2 \end{array} \right] \end{array} \right\} & 1 \\
 700 & & \\
 800 & & \\
 \hline
 900 & & \\
 1000 & & \\
 1100 & & \\
 1200 & &
 \end{array}$$

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